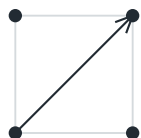


Move from dot to neighboring dot: one step right R, up U, or diagonally right-and-up D. R and U each cost 1 coin. D costs the amount on its price card. These are prices, not string lengths.



R then U: 2 coins

D: its card price

Problem 1: For each price card, build the cheapest route from S to F. Do all cheapest routes use the same number of diagonal steps? Explain how the price changes your choice.



D costs 1

Cheapest:

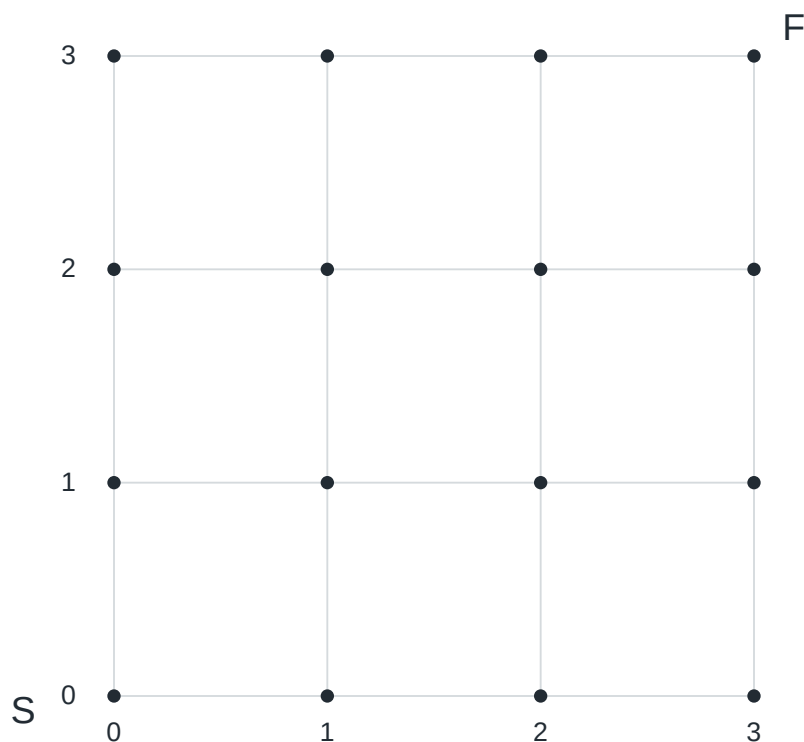
D costs 2

Cheapest:

D costs 3

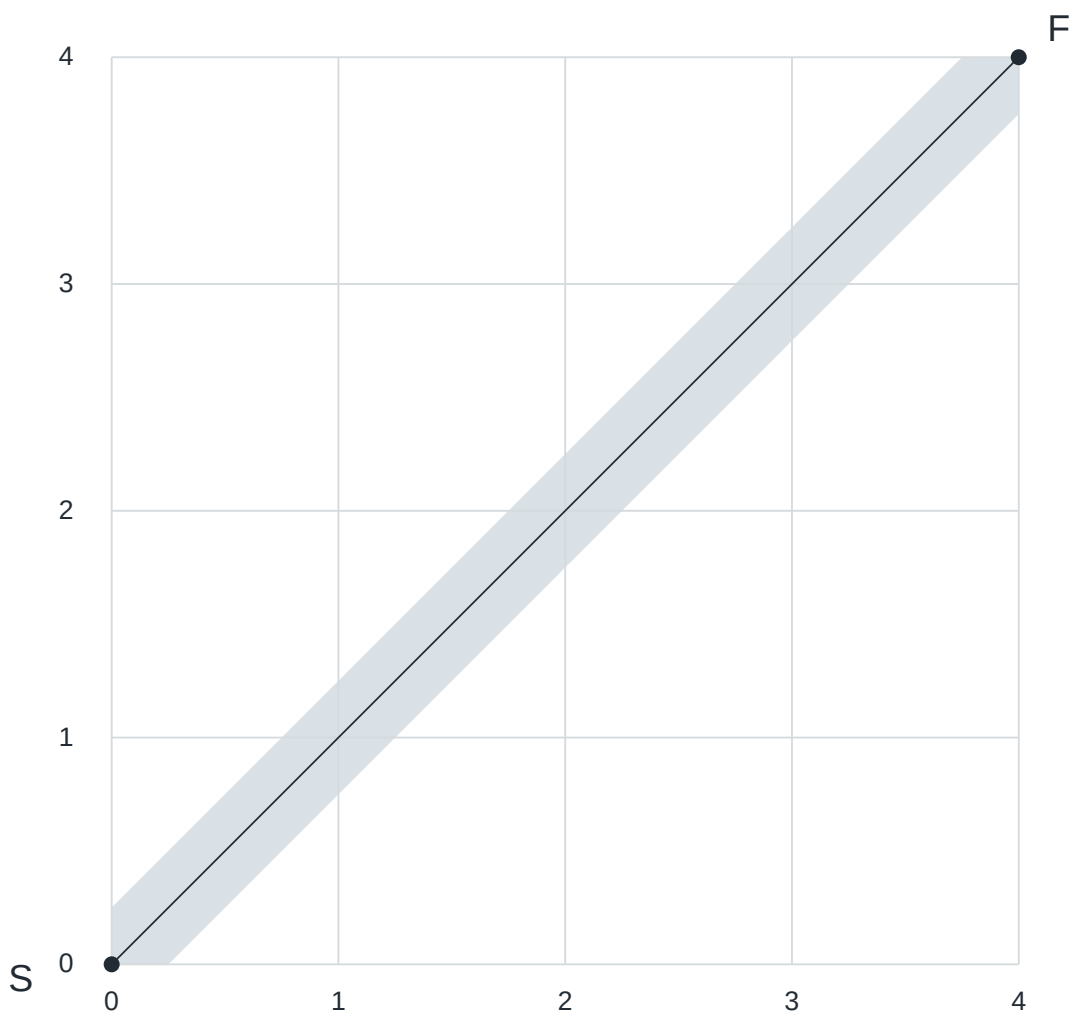
Cheapest:

Problem 2: Use only right and up steps along grid lines. Find every S-to-F route. Then block one dot other than S or F: routes cannot pass through it. Find a blocked dot that leaves exactly 8 routes, and one that leaves exactly 11. Could one blocked dot stop every route? Explain how you check your counts.



A route can be recorded as a word, such as RRUURU.

Problem 3: Stay inside the gray strip, including its boundary, using only horizontal and vertical pieces. Left and down are allowed. Retracing a piece counts its full length again. Make an S-to-F route of length 30 units, then one of length 100 units. Is there a longest finite route? A record may give a piece and its repeat count.



Gray strip: at most $\frac{1}{4}$ unit vertically from the diagonal.


 1 unit = 30 mm